

2. W drugim oknie wykonanie ping 8.8.8.8 oraz curl https://example.com

ping 8.8.8.8 : (To polecenie wygeneruje pakiety ICMP Echo Request oraz ICMP Echo Reply)

```
kali@kali: ~  
Session Actions Edit View Help  
~(kali@kali)-[~]  
$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=12.1 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=13.0 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=12.2 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=11.5 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=64 time=11.8 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=64 time=11.9 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=64 time=11.3 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=64 time=11.6 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=64 time=11.0 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=64 time=10.9 ms  
64 bytes from 8.8.8.8: icmp_seq=11 ttl=64 time=12.4 ms  
64 bytes from 8.8.8.8: icmp_seq=12 ttl=64 time=11.0 ms  
64 bytes from 8.8.8.8: icmp_seq=13 ttl=64 time=10.7 ms  
64 bytes from 8.8.8.8: icmp_seq=14 ttl=64 time=11.3 ms  
64 bytes from 8.8.8.8: icmp_seq=15 ttl=64 time=11.7 ms  
64 bytes from 8.8.8.8: icmp_seq=16 ttl=64 time=10.7 ms  
64 bytes from 8.8.8.8: icmp_seq=17 ttl=64 time=11.3 ms  
64 bytes from 8.8.8.8: icmp_seq=18 ttl=64 time=11.1 ms  
64 bytes from 8.8.8.8: icmp_seq=19 ttl=64 time=10.7 ms  
64 bytes from 8.8.8.8: icmp_seq=20 ttl=64 time=11.7 ms  
64 bytes from 8.8.8.8: icmp_seq=21 ttl=64 time=10.9 ms  
64 bytes from 8.8.8.8: icmp_seq=22 ttl=64 time=12.0 ms
```

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
113	53.106210401	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=54/13824, ttl=64 (reply in 114)
114	53.118472606	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=54/13824, ttl=64 (request in 113)
115	54.108413052	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=55/14080, ttl=64 (reply in 116)
116	54.119528941	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=55/14080, ttl=64 (request in 115)
117	55.109490645	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=56/14336, ttl=64 (reply in 118)
118	55.121197196	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=56/14336, ttl=64 (request in 117)
119	56.110940931	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=57/14592, ttl=64 (reply in 120)
120	56.122366978	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=57/14592, ttl=64 (request in 119)
121	57.112350190	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=58/14848, ttl=64 (reply in 122)
122	57.123852762	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=58/14848, ttl=64 (request in 121)
123	58.113650759	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=59/15104, ttl=64 (reply in 124)
124	58.125312472	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=59/15104, ttl=64 (request in 123)
125	59.115711431	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=60/15360, ttl=64 (reply in 126)
126	59.126455133	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=60/15360, ttl=64 (request in 125)
127	60.119926695	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=61/15616, ttl=64 (reply in 128)
128	60.131958120	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=61/15616, ttl=64 (request in 127)
129	61.120563431	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=62/15872, ttl=64 (reply in 130)
130	61.132121681	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=62/15872, ttl=64 (request in 129)
131	62.122604552	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request id=0xc07e, seq=63/16128, ttl=64 (no response found!)
132	62.133320204	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply id=0xc07e, seq=63/16128, ttl=64 (request in 131)

Frame 1: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0

Ethernet II, Src: PCSystemtec 5a:87:bc (08:00:27:5a:87:bc), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (request)

0000 ff ff ff ff ff ff 00 27 5a 87 bc 08 06 00 01 'Z'....
0010 08 00 06 04 00 01 08 00 27 5a 87 bc 0a 00 02 0f 'Z'....
0020 00 00 00 00 00 00 0a 00 02 02

Po przerwaniu operacji:

```
^C  
— 8.8.8.8 ping statistics —  
198 packets transmitted, 198 received, 0% packet loss, time 197250ms  
rtt min/avg/max/mdev = 10.597/11.472/13.041/0.525 ms  
~(kali@kali)-[~]  
$
```

curl <https://example.com> : Narzędzie curl pobierze w tle kod źródłowy tej strony example.om, który jest przykładem do naszego ćwiczenia. W konsoli wyświetli się czysty kod HTML.

```
(kali@kali)-[~]
$ curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:sans-serif}h1{font-size:1.5em}div[opacity:0.8]{a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>

(kali@kali)-[~]
```

W Wireshark otrzymamy wiele danych zwrotnych:

No.	Time	Source	Destination	Protocol	Length	Info
4	0.011199623	fd17:625c:f037:12:d5...	2606:4700:10::ac42:...	TCP	94	47400 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=3041065498 TSecr=0 WS=1024
5	0.011963952	2606:4700:10::ac42:...	fd17:625c:f037:2:d5...	TCP	74	443 → 47400 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.011963217	192.168.1.1	10.0.2.15	DNS	103	Standard query response 0x576e A example.com A 172.66.147.243 A 104.20.23.154
7	0.012172190	fd17:625c:f037:2:d5...	2606:4700:10::6814:...	TCP	94	55774 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=2291886170 TSecr=0 WS=1024
8	0.012631040	2606:4700:10::6814:...	fd17:625c:f037:2:d5...	TCP	74	443 → 55774 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
9	0.012793776	10.0.2.15	172.66.147.243	TCP	74	51476 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1400 SACK_PERM TSval=2115104190 TSecr=0 WS=1024
10	0.024580844	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400
11	0.024667129	10.0.2.15	172.66.147.243	TCP	54	51476 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
12	0.025716800	10.0.2.15	172.66.147.243	TLShv1.3	1749	Client Hello (SNI=example.com)
13	0.026034925	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [ACK] Seq=1 Ack=1461 Win=65535 Len=0
14	0.026035090	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [ACK] Seq=1 Ack=1696 Win=65535 Len=0
15	0.037002906	172.66.147.243	10.0.2.15	TLShv1.3	1494	Server Hello, Change Cipher Spec
16	0.037003257	172.66.147.243	10.0.2.15	TCP	1494	443 → 51476 [ACK] Seq=1441 Ack=1696 Win=65535 Len=1440 [TCP PDU reassembled in 20]
17	0.037065928	10.0.2.15	172.66.147.243	TCP	54	51476 → 443 [ACK] Seq=1696 Ack=1441 Win=65535 Len=0
18	0.037160295	10.0.2.15	172.66.147.243	TCP	54	51476 → 443 [ACK] Seq=1696 Ack=2881 Win=65535 Len=0
19	0.037364415	172.66.147.243	10.0.2.15	TCP	1494	443 → 51476 [ACK] Seq=2881 Ack=1696 Win=65535 Len=1440 [TCP PDU reassembled in 20]
20	0.037364578	172.66.147.243	10.0.2.15	TLShv1.3	813	Application Data
21	0.037371885	10.0.2.15	172.66.147.243	TCP	54	51476 → 443 [ACK] Seq=1696 Ack=4321 Win=65535 Len=0
22	0.037406030	10.0.2.15	172.66.147.243	TCP	54	51476 → 443 [ACK] Seq=1696 Ack=5080 Win=65535 Len=0
23	0.038979005	10.0.2.15	172.66.147.243	TLShv1.3	134	Change Cipher Spec, Application Data
24	0.039211115	10.0.2.15	172.66.147.243	TLShv1.3	177	Application Data
25	0.039702428	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [ACK] Seq=5080 Ack=1776 Win=65535 Len=0
26	0.039702676	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [ACK] Seq=5080 Ack=1899 Win=65535 Len=0
27	0.039714132	10.0.2.15	172.66.147.243	TLShv1.3	102	Application Data
28	0.039985182	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [ACK] Seq=5080 Ack=1947 Win=65535 Len=0
29	0.048814556	172.66.147.243	10.0.2.15	TLShv1.3	598	Application Data, Application Data
30	0.049035704	10.0.2.15	172.66.147.243	TLShv1.3	85	Application Data
31	0.049087755	172.66.147.243	10.0.2.15	TLShv1.3	85	Application Data
32	0.052943207	172.66.147.243	10.0.2.15	TLShv1.3	833	Application Data, Application Data
33	0.053049652	10.0.2.15	172.66.147.243	TCP	54	51476 → 443 [ACK] Seq=1978 Ack=6434 Win=65535 Len=0
34	0.053215800	10.0.2.15	172.66.147.243	TLShv1.3	102	Application Data
35	0.053416988	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [ACK] Seq=6434 Ack=2026 Win=65535 Len=0
36	0.053424213	10.0.2.15	172.66.147.243	TLShv1.3	78	Application Data
37	0.053713594	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [ACK] Seq=6434 Ack=2050 Win=65535 Len=0
38	0.053803947	10.0.2.15	172.66.147.243	TCP	54	51476 → 443 [FIN, ACK] Seq=2050 Ack=6434 Win=65535 Len=0
39	0.054010261	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [ACK] Seq=6434 Ack=2051 Win=65535 Len=0
40	0.060309955	172.66.147.243	10.0.2.15	TCP	60	443 → 51476 [FIN, ACK] Seq=6434 Ack=2051 Win=65535 Len=0
41	0.060330939	10.0.2.15	172.66.147.243	TCP	54	51476 → 443 [ACK] Seq=2051 Ack=6435 Win=15104 Len=0
42	5.040272150	fe80::0af1:c892:ee1...	fe80::2	ICMPv6	80	Neighbor Solicitation for fe80::2 from 08:00:27:5a:87:bc
43	5.040505146	fe80::2	fe80::0af1:c892:ee1...	ICMPv6	80	Neighbor Advertisement fe80::2 (rttr, sol, ovr) is at 52:54:00:12:35:00

3. Zatrzymanie capture, filtrowanie po icmp — identyfikacja warstw OSI w jednym pakiecie ICMP:

W celu zatrzymania klikam na czerwony przycisk "stop capturing packets"

Filtruje po protokole ICMP

Klikam dwukrotnie na pakiet numer 1 protokołu icmp (Echo ping)

```
Wireshark · Packet 1 · eth0
▶ Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0
▶ Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00)
▶ Internet Protocol Version 4, Src: 10.0.2.15, Dst: 8.8.8.8
▶ Internet Control Message Protocol

0000  52 54 00 12 35 00 08 00 27 5a 87 bc 08 00 45 00  RT..5... 'Z...E.
0010  00 54 17 77 40 00 40 01 07 14 0a 00 02 0f 08 08  -T·w@·@· .....
0020  08 08 08 00 fd 95 c0 7f 00 01 3d bc 7d 6a 00 00  ..... ..=.}j..
0030  00 00 b9 ef 06 00 00 00 00 00 10 11 12 13 14 15  .....
0040  16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25  ..... !"#$$%
0050  26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35  &'()*+,- ./012345
0060  36 37                                           67
```

Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0 -
WARSTWA 1 – FIZYCZNA (identyfikuje rozmiar ramki 98 bajtów - 784 bity)

```
Wireshark · Packet 1 · eth0
▼ Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0
  Section number: 1
  ▶ Interface id: 0 (eth0)
    Encapsulation type: Ethernet (1)
    Arrival Time: Aug 13, 2026 08:44:45.454603437 EDT
    UTC Arrival Time: Aug 13, 2026 12:44:45.454603437 UTC
    Epoch Arrival Time: 1786625085.454603437
    [Time shift for this packet: 0.000000000 seconds]
    [Time since reference or first frame: 0.000000000 seconds]
    Frame Number: 1
    Frame Length: 98 bytes (784 bits)
    Capture Length: 98 bytes (784 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:icmp:data]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ICMP]
    [Coloring Rule String: icmp || icmpv6]
```

Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00) -
WARSTWA 2 - ŁĄCZA DANYCH (identyfikuje źródłowy adres MAC i docelowy adres MAC)

```
▼ Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00)
  ▼ Destination: 52:54:00:12:35:00 (52:54:00:12:35:00)
    .... ..1. .... .. = LG bit: Locally administered address (this is NOT the factory default)
    .... ..0. .... .. = IG bit: Individual address (unicast)
  ▼ Source: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc)
    .... ..0. .... .. = LG bit: Globally unique address (factory default)
    .... ..0. .... .. = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
```

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 8.8.8.8 - **WARSTWA 3 – SIECIOWA** (identyfikuje adres IP źródłowy oraz adres IP docelowy)

```
▼ Internet Protocol Version 4, Src: 10.0.2.15, Dst: 8.8.8.8
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x1777 (6007)
  ▶ 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: ICMP (1)
    Header Checksum: 0x0714 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.0.2.15
    Destination Address: 8.8.8.8
    [Stream index: 0]
  ▶ Internet Control Message Protocol

0000  52 54 00 12 35 00 08 00 27 5a 87 bc 08 00 45 00  RT...5... 'Z....E.
0010  00 54 17 77 40 00 40 01 07 14 0a 00 02 0f 08 08  .T.w@.@. ....
0020  08 08 08 00 fd 95 c0 7f 00 01 3d bc 7d 6a 00 00  .. ...=.}j..
0030  00 00 b9 ef 06 00 00 00 00 00 10 11 12 13 14 15  .. ...
0040  16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25  .. ... !"#$$%
0050  26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35  &'()*+,-./012345
0060  36 37 67
```

Internet Control Message Protocol - **WARSTWA 3 – SIECIOWA**

Type: Informuje, czy to żądanie (Type: 8 - Echo request), czy odpowiedź (Type: 0 - Echo reply)

Code: Dodatkowy kod diagnostyczny (dla udanego pinga wynosi 0)

```
▼ Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0xfd95 [correct]
  [Checksum Status: Good]
  Identifier (BE): 49279 (0xc07f)
  Identifier (LE): 32704 (0x7fc0)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Response frame: 2]
  ▶ ICMP Data: 3dbc7d6a00000000b9ef06000000000101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637

0000  52 54 00 12 35 00 08 00 27 5a 87 bc 08 00 45 00  RT...5... 'Z....E.
0010  00 54 17 77 40 00 40 01 07 14 0a 00 02 0f 08 08  .T.w@.@. ....
0020  08 08 08 00 fd 95 c0 7f 00 01 3d bc 7d 6a 00 00  .. ...=.}j..
0030  00 00 b9 ef 06 00 00 00 00 00 10 11 12 13 14 15  .. ...
0040  16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25  .. ... !"#$$%
0050  26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35  &'()*+,-./012345
0060  36 37 67
```

4. Filtrowanie po tcp.port == 443 — obserwacja TLS handshake

Wpisanie w pasku wyszukiwania frazy: tcp.port == 443

Wireshark capture of a TLS handshake packet (Frame 10). The packet is filtered by 'tcp.port == 443'. The packet details show it's a TLSv1.3 Client Hello. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Frame 10: Packet, 1749 bytes on wire (13992 bits), 1749 bytes captured (13992 bits) on interface eth0, id 0

Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 172.66.147.243

Transmission Control Protocol, Src Port: 33636, Dst Port: 443, Seq: 1, Ack: 1, Len: 1695

Transport Layer Security

[Stream index: 0]

TLSv1.3 Record Layer: Handshake Protocol: Client Hello

Content Type: Handshake (22)

Version: TLS 1.0 (0x0301)

Length: 1690

Handshake Protocol: Client Hello

Handshake Type: Client Hello (1)

Length: 1686

Version: TLS 1.2 (0x0303)

Random: 4f8f44452fc0ce0077c2dc63dc9f97ff5147e43990efaea2ef730a506fcb56fe

Session ID Length: 32

Session ID: 09789eef7f4cb317f5017d7e23fc70a3ccbc3f9b238d5d978e2ec63c5c789a6

Cipher Suites Length: 180

Cipher Suites (90 suites)

Compression Methods Length: 1

Compression Methods (1 method)

0000 52 54 00 12 35 00 08 00 27 5a 87 bc 08 00 45 00 RT - 5 - - - - E

0010 06 c7 33 22 40 00 00 00 b4 ca 0a 00 02 0f ac 42 - 3 "g @ - - - - B

0020 93 f3 03 64 01 bb b6 39 3e 23 92 dd 66 02 50 18 - d - - 9 ># - - f P

0030 fa f0 52 fe 00 00 16 03 01 06 9a 01 00 06 96 03 - R - - - - - - - -

0040 03 4f 8f 44 45 2f c0 ce 00 77 c2 dc 63 dc 9f 97 - 0 DE / - - w - c -

0050 ff 51 47 e4 39 90 ef ae a2 ef 73 0a 50 6f cb 56 - QG 9 - - - s Po V

0060 fe 20 09 78 9e ef 7f 4c b3 17 f5 01 7d 7e 23 fc - x - - L - - - } - #

0070 70 a3 cc bc a3 f9 b2 38 d5 d9 78 e2 ec 63 c5 c7 p - - - - 8 - - x - - c -

0080 89 a6 00 b4 13 02 13 03 13 01 c0 2c c0 30 00 a3 - - - - - - - - - - -

0090 00 9f cc a9 cc a8 cc aa c0 ad c0 9f c0 5d c0 61 - - - - - - - - - -] a

00a0 c0 57 c0 53 00 a7 c0 2b c0 2f 00 a2 00 9e c0 ac - W S - - - + / - - - -

00b0 c0 9e c0 5c c0 60 c0 56 c0 52 00 a6 c0 af c0 ae - \ - - \ - V - R - - -

00c0 c0 a3 c0 a2 c0 24 c0 28 00 6b 00 6a c0 73 c0 77 - - - - \$ (- k j s w

00d0 00 c4 00 c3 00 6d 00 c5 c0 23 c0 27 00 67 00 40 - - - - m - - - ' g @

00e0 c0 72 c0 76 00 b6 00 bd 00 6c 00 bf c0 0a c0 14 - r v - - - - l - - - -

00f0 00 39 00 38 00 38 00 87 c0 19 00 3a 00 89 c0 09 - 9 8 - - - - - - - -

0100 c0 13 00 33 00 32 00 9a 00 99 00 45 00 44 c0 18 - 3 2 - - - - E D -

0110 00 34 00 9b 00 46 00 9d c0 9d c0 51 00 9c c0 9c - 4 - - F - - - Q - - -

0120 c0 50 c0 a1 c0 a0 00 3d 00 c0 00 3c 00 ba 00 35 - P - - - = - - - < - 5

0130 00 84 00 2f 00 96 00 41 01 00 05 99 ff 01 00 01 - - / - - A - - - - -

0140 00 00 00 00 10 00 0e 00 00 0b 65 78 61 6d 70 6c - - - - - - - - - - exampl

No. 10 - Time: 0.033808048 - Source: 10.0.2.15 - Destination: 172.66.147.243 - Protocol: TLSv1.3 - Length: 1749 - Info: Client Hello (SNI=example.com)

Show packet bytes Layout: Vertical (Stacked)

Obserwacja TLS handshake: (klient się wita i zgłasza metody szyfrowania)

Wireshark capture of a TLS handshake packet (Frame 10). The packet is filtered by 'tcp.port == 443'. The packet details show it's a TLSv1.3 Client Hello. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Frame 10: Packet, 1749 bytes on wire (13992 bits), 1749 bytes captured (13992 bits) on interface eth0, id 0

Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 172.66.147.243

Transmission Control Protocol, Src Port: 33636, Dst Port: 443, Seq: 1, Ack: 1, Len: 1695

Transport Layer Security

[Stream index: 0]

TLSv1.3 Record Layer: Handshake Protocol: Client Hello

Content Type: Handshake (22)

Version: TLS 1.0 (0x0301)

Length: 1690

Handshake Protocol: Client Hello

Handshake Type: Client Hello (1)

Length: 1686

Version: TLS 1.2 (0x0303)

Random: 4f8f44452fc0ce0077c2dc63dc9f97ff5147e43990efaea2ef730a506fcb56fe

Session ID Length: 32

Session ID: 09789eef7f4cb317f5017d7e23fc70a3ccbc3f9b238d5d978e2ec63c5c789a6

Cipher Suites Length: 180

Cipher Suites (90 suites)

Compression Methods Length: 1

Compression Methods (1 method)

Extensions Length: 1433

Extension: renegotiation_info (len=1)

Extension: server_name (len=16) name=example.com

Extension: ec_point_formats (len=2)

Extension: supported_groups (len=18)

Extension: application_layer_protocol_negotiation (len=14)

Extension: encrypt_then_mac (len=0)

Extension: extended_master_secret (len=0)

Extension: post_handshake_auth (len=0)

Extension: signature_algorithms (len=60)

Extension: supported_versions (len=5) TLS 1.3, TLS 1.2

Extension: psk_key_exchange_modes (len=2)

Extension: key_share (len=1258) X25519MLKEM768, x25519

Extension: compress_certificate (len=5)

[JA4: t13d9013h2_c6771aded2ed_57a60bdf03d1]

[JA4_r [...]: t13d9013h2_002f,0032,0033,0034,0035,0038,0039,003a,003c,003d,0040,0041,0044,0045,0046,0067,006a,006b,006c,006d,0084,0087,0088,0089,0096,0099,009a,

[JA3 Fullstring [...]: 771,4866-4867-4865-49196-49200-163-159-52393-52392-52394-49325-49311-49245-49249-49239-49235-167-49195-49199-162-158-49324-49310-49244-49

[JA3: c654189f0c9cbf638bb74b824e790138]

0130 00 84 00 2f 00 96 00 41 01 00 05 99 ff 01 00 01 - - / - - A - - - - -

Length of hello extensions (tls.handshake.extensions_length), 2 bytes

Show packet bytes Layout: Vertical (Stacked)

Serwer akceptuje (odpowiedź serwera, serwer przesyła swój certyfikat TLS):

Wireshark - Packet 13 - eth0

- Frame 13: Packet, 1494 bytes on wire (11952 bits), 1494 bytes captured (11952 bits) on interface eth0, id 0
- Ethernet II, Src: 52:54:00:12:35:00 (52:54:00:12:35:00), Dst: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc)
- Internet Protocol Version 4, Src: 172.66.147.243, Dst: 10.0.2.15
- Transmission Control Protocol, Src Port: 443, Dst Port: 33636, Seq: 1, Ack: 1696, Len: 1440
- Transport Layer Security
 - [Stream index: 0]
 - TLSv1.3 Record Layer: Handshake Protocol: Server Hello
 - Content Type: Handshake (22)
 - Version: TLS 1.2 (0x0303)
 - Length: 1210
 - Handshake Protocol: Server Hello
 - Handshake Type: Server Hello (2)
 - Length: 1206
 - Version: TLS 1.2 (0x0303)
 - [Expert Info (Chat/Deprecated): This legacy version field MUST be ignored. The supported_versions extension is present and MUST be used instead.]
 - [This legacy version field MUST be ignored. The supported_versions extension is present and MUST be used instead.]
 - [Severity level: Chat]
 - [Group: Deprecated]
 - Random: eadffe47c078b93ae1fddiffa6527bcffc41558004c931ab7786fbb62bef3152
 - Session ID Length: 32
 - Session ID: 09789eef7f4cb317f5017d7e23fc70a3ccbca3f9b238d5d978e2ec63c5c789a6
 - Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)
 - Compression Method: null (0)
 - Extensions Length: 1134
 - Extension: key_share (len=1124) X25519MLKEM768
 - Type: key_share (51)
 - Length: 1124
 - Key Share extension
 - Extension: supported_versions (len=2) TLS 1.3
 - Type: supported_versions (43)
 - Length: 2
 - Supported Version: TLS 1.3 (0x0304)
 - [JA3S Fullstring: 771,4866,51-43]
 - [JA3S: 907bf3ecef1c987c889946b737b43de8]
 - TLSv1.3 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
 - Content Type: Change Cipher Spec (20)
 - Version: TLS 1.2 (0x0303)
 - Length: 1
 - Change Cipher Spec Message
 - TLS segment data (219 bytes)

0000 08 00 27 5a 87 bc 52 54 00 12 35 00 08 00 45 00 ..'Z'..RT..5...E.....

No.: 13 - Time: 0.047846134 - Source: 172.66.147.243 - Destination: 10.0.2.15 - Protocol: TLSv1.3 - Length: 1494 - Info: Server Hello, Change Cipher Spec

✓ Show packet bytes Layout: Vertical (Stacked)

Close Help